

Graph Theory

How to turn pictures, roads, and friendships into mathematics.

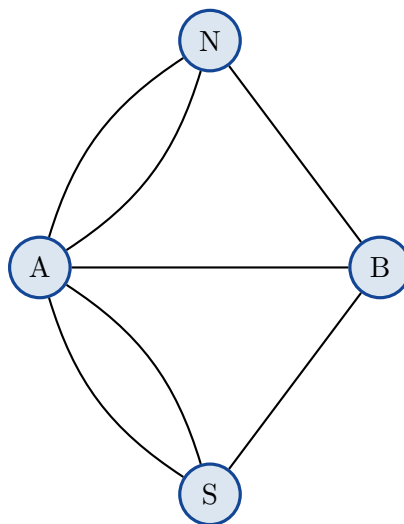
BdMO Camp 2026

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1. The Birth of Graph Theory

In 1736, Leonhard Euler solved a puzzle about the city of Königsberg: *Is it possible to walk through the city and cross each of its seven bridges exactly once?*

Euler realized that the exact shape of the islands didn't matter. The only thing that mattered was **what was connected to what**. He replaced the landmasses with dots (vertices) and the bridges with lines (edges).



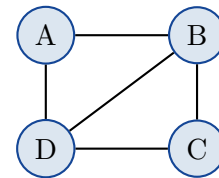
The Königsberg Bridge Graph: 4 landmasses and 7 bridges.

2. The Dictionary: What is a Graph?

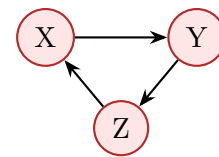
A graph $G = (V, E)$ is a collection of vertices (dots) and edges (lines connecting them).

Basic Terminology

- **Vertex:** The dots (e.g., A, B, C).
- **Edge:** The lines connecting the dots. In an **undirected graph**, these are two-way connections.
- **Directed Graph:** Edges have a specific direction (like one-way streets), drawn as arrows.
- **Degree $\deg(v)$:** Number of edges at a vertex. In directed graphs, we split this into **in-degree** (arrows pointing in) and **out-degree** (arrows pointing out).
- **Path:** A sequence of edges without repeating vertices.
- **Cycle:** A path that starts and ends at the same vertex.



Undirected Graph



Directed Cycle (Out-degree = 1)

3. The First Theorem: Handshaking Lemma

Imagine every edge is a physical string connecting two dots. Every string has exactly two ends. If you count the number of string ends attached to each dot (this is the **degree**) and add them all up, you are simply counting all the string ends in the entire graph. Because every string has exactly 2 ends, the sum must be $2 \times$ (number of strings).

This simple observation is the foundation of graph theory counting!

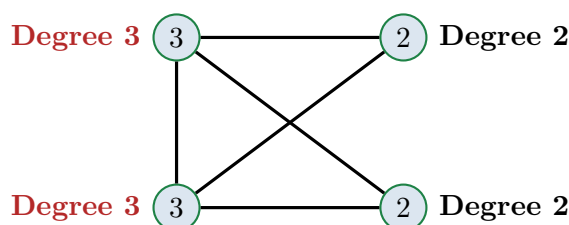
The Handshaking Lemma

For any graph, the sum of the degrees of all vertices is exactly twice the number of edges:

$$\sum_{v \in V} \deg(v) = 2|E|$$

The Parity Consequence: Since $2|E|$ is always an even number, the sum of all degrees *must* be even. Try adding three odd numbers together—you get an odd number, which is impossible here! To keep the total sum even, you can only ever have an even number of odd numbers. Therefore:

In any graph, the number of vertices with an odd degree must be even.



Degrees: 3, 3, 2, 2. The sum is 10 (which is 2×5 edges). Notice there are exactly **two** odd-degree vertices.

4. Eulerian Paths and Cycles

Returning to Euler's original bridge puzzle, we want to know when we can trace a graph without lifting our pen and without repeating any lines.

Think about the intuition: every time your pencil **enters** a vertex, it must also **leave** that vertex to continue the drawing. This means the edges connected to any "middle" vertex on your journey must come in pairs (one path in, one path out). Therefore, any vertex you pass through must have an **even** degree.

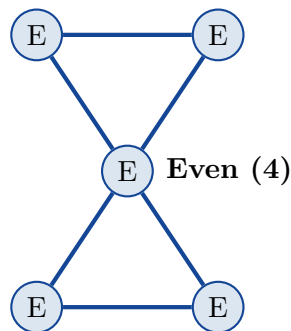
The only exceptions are the vertex where you **start** your drawing (you leave it, but didn't enter) and the vertex where you **finish** (you enter it, but don't leave).

Euler's Criteria

For a connected graph:

- **Eulerian Cycle** (crosses every edge once and returns to the start) exists **if and only if** every single vertex has an **even degree**.
- **Eulerian Path** (crosses every edge once but starts and ends at different places) exists **if and only if** exactly **0 or 2** vertices have an **odd degree**.

Even (2) Even (2)



Even (2) Even (2)

Because every single vertex has an even degree, we can trace this entire figure without lifting our pen, and we will finish exactly where we started (an Eulerian Cycle).

5. Bipartite Graphs

Imagine a graph representing a group of workers and a list of jobs. Edges connect workers to the jobs they are qualified for. You will never see an edge connecting a worker to another worker, or a job to another job. This is called a bipartite graph.

Definition

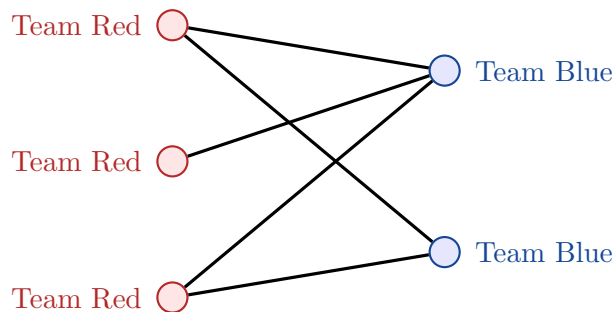
A graph is **bipartite** if its vertices can be split into two distinct teams (e.g., Team Left and Team Right) such that **every edge connects a Left vertex to a Right vertex**. Vertices on the same team are never connected.

A great way to test if a graph is bipartite is by coloring it. Start at any vertex and color it **Red**. Because it cannot connect to its own team, all of its neighbors must be **Blue**. Their neighbors must be **Red**, and so on.

If a graph has a cycle with an *odd* number of vertices (like a triangle), this alternating Red-Blue-Red coloring will fail when you return to the start, forcing two vertices of the same color to connect. Because of this, bipartite graphs have a very strict rule:

The Bipartite Condition

A graph is bipartite **if and only if** it contains absolutely no cycles of an **odd length**.



Notice how all edges only cross the gap. There are no triangles, pentagons, or any odd cycles.

6. Trees

In graph theory, a "tree" is the ultimate minimalist structure. It is a connected graph with absolutely zero redundancy.

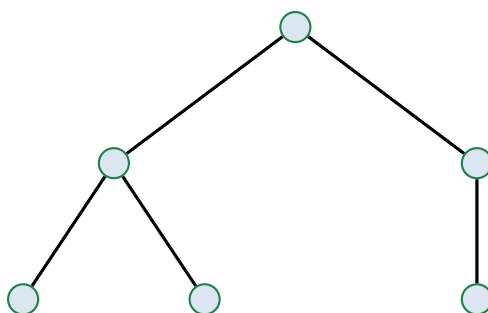
You can think of a cycle in a graph as a "redundant" path—a second way to get from point A to point B. Trees do not have these. They represent the bare minimum number of edges required to keep a set of vertices connected. Because they have no redundant edges, they follow very strict mathematical rules.

Properties of Trees

A **tree** is a connected graph with no cycles. For any graph with exactly n vertices, the following statements mean the exact same thing:

1. The graph is a tree.
2. The graph is fully connected and has exactly $n - 1$ edges.
3. There is exactly one unique path between any two vertices.
4. Removing any single edge will instantly break the graph into disconnected pieces.

In Olympiad problems, if you can prove a graph has $n - 1$ edges and is connected, you get to use all the other facts (like having no cycles) for free!

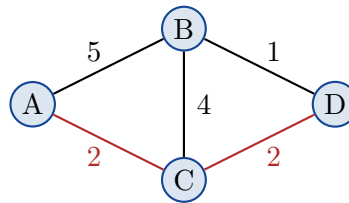


A tree with $n = 6$ vertices. Notice it has exactly 5 edges. If you delete any edge, the graph falls apart.

7. Algorithmic Applications & Advanced Modeling

A. Shortest Path (Weighted Graphs)

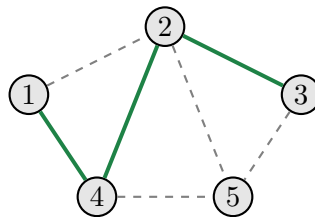
If edges represent roads, we assign "weights" (distances) to them. Algorithms find the absolute fastest route from node A to node B.



The shortest path from A to D is $A \rightarrow C \rightarrow D$ (Total distance: 4).

B. Minimum Spanning Trees (MST)

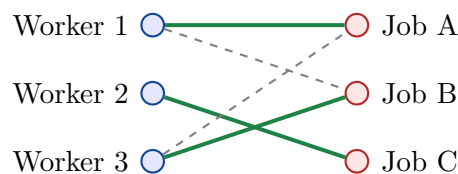
How do you connect a network using the least amount of expensive cable? You find the Minimum Spanning Tree—a subgraph that connects all vertices without cycles and with the lowest total edge weight.



The green lines form a Spanning Tree that connects all nodes with minimum cost.

C. Bipartite Matching & Network Flow

If you have 3 workers and 3 jobs, but each worker is only qualified for certain jobs, can you assign every job? This is a **Bipartite Matching** problem.



A Perfect Matching (green lines): every worker is assigned exactly one job they can do.

8. BdMO Problem Session

Work on these problems using the space provided. If you get stuck, flip to the next page for hints!

Problem 1: The Teamwork Schedule

At the BdMO camp, there is an odd number of students. To build teamwork, the organizers want every student to pair up for mini-projects with *exactly half* of the other students present. Is it possible to create this schedule if there are 9 students? What about 11 students? What about 25 students?

Hints for Problem 1

- **Hint 1:** Translate this to graph theory. What do the vertices represent? What do the edges represent?
- **Hint 2:** If there are n students, how many "other students" are there?
- **Hint 3:** What is the exact degree of every single vertex if they shake hands with *half* of the others? Calculate this number for $n = 9$, $n = 11$, and $n = 25$.
- **Hint 4:** Apply the Handshaking Lemma. What does the lemma say about the number of vertices that are allowed to have an odd degree?

Problem 2: The Alien Council

An alien ambassador meets with a council of 6 human leaders. Prove that among these 6 leaders, there must be a group of 3 who can all converse directly with each other in a shared language, OR a group of 3 where no two people share any common language at all.

Hints for Problem 2

- **Hint 1:** Draw 6 vertices representing the leaders. Draw an edge between every single pair of leaders (this is called a complete graph, K_6).
- **Hint 2:** Color the edges! Make an edge **RED** if they share a language, and **BLUE** if they don't.
- **Hint 3:** Pick one specific leader (Vertex A). There are 5 edges connected to A. By the Pigeonhole Principle, at least how many of these 5 edges **MUST** be the exact same color?
- **Hint 4:** Assume A has 3 red edges connecting to leaders B, C, and D. What happens if B and C share a language (a red edge)? What happens if none of B, C, or D share a language (all blue edges)?

Problem 3: The Chaotic Cleanup

During a water balloon fight, every student throws exactly one balloon at exactly one other person. Prove that it is always possible to divide the students into 3 cleanup teams so that no student is on the same team as the person they threw their balloon at.

Hints for Problem 3

- **Hint 1:** Model this as a **directed graph**. A directed edge points from the thrower to the target.
- **Hint 2:** What is the **out-degree** (number of outgoing arrows) of every single vertex in this graph?
- **Hint 3:** If every vertex has an out-degree of exactly 1, what does the graph look like? Start drawing random examples. You will notice it always forms directed cycles with "trees" branching into them.
- **Hint 4:** You are trying to assign 3 "colors" (teams) to the vertices so that no two connected vertices have the same color. Can you always 3-color any cycle, regardless of whether its length is even or odd?

Problem 4: The Robotic Vacuum

A robotic vacuum is placed on a 5×5 grid of square tiles. Every minute, it moves to exactly one adjacent tile (up, down, left, or right). Can the robot clean all the tiles and return to its starting position in exactly 25 minutes?

Hints for Problem 4

- **Hint 1:** The tiles are vertices, and the legal moves between adjacent tiles are edges.
- **Hint 2:** Imagine the 5×5 grid is colored like a chessboard (black and white tiles). When the robot moves up, down, left, or right, what happens to the color of the tile it is on?
- **Hint 3:** Because it always alternates colors, this graph is **bipartite**.
- **Hint 4:** The robot wants to return to its starting position. In graph theory terms, this is a **cycle**. What does the Bipartite Theorem tell us about the length of cycles in a bipartite graph? Is 25 an allowed length?

Problem 5: Minimum Roads (Challenge)

A country has n cities. The government builds two-way roads between them so that it is possible to travel between any two cities. To save money, they build the absolute minimum number of roads required to keep the country connected. Prove that there must be at least two cities that have only one road connected to them.

Hints for Problem 5

- **Hint 1:** A connected graph with the "absolute minimum number of edges" is the definition of a **tree**.
- **Hint 2:** A city with only one road connected to it has a degree of 1. In a tree, this is called a **leaf**. You are trying to prove every tree has at least 2 leaves.
- **Hint 3:** How many edges does a tree with n vertices have? ($|E| = n - 1$).
- **Hint 4:** Use the Handshaking Lemma! The sum of all degrees is $2|E|$, which means $\sum \deg(v) = 2(n - 1) = 2n - 2$.
- **Hint 5:** What would happen if the statement was false, and every city had 2 or more roads? What would the minimum sum of degrees be? Compare that to $2n - 2$.

Problem 6: The Art Gallery

An art gallery has 12 interconnected rooms. Every single room has exactly 3 doors leading to other rooms. Is it possible for a security guard to walk through the gallery, passing through every single door exactly once?

Hints for Problem 6

- **Hint 1:** The rooms are vertices and the doors are edges.
- **Hint 2:** Passing through every edge exactly once means we are looking for an **Eulerian Path** or an **Eulerian Cycle**.
- **Hint 3:** What is the degree of every vertex in this graph? How many vertices are there?
- **Hint 4:** Look at Euler's Criteria in Section 4 of your notes. How many odd-degree vertices are allowed if an Eulerian Path is going to exist? Does this graph pass or fail the test?

Problem 7: Triangle-Free Construction

Is it possible to draw a graph with 7 vertices and 10 edges that does *not* contain a triangle (a cycle of length 3)? If yes, try to draw it!

Hints for Problem 7

- **Hint 1:** A triangle is an odd cycle (C_3). What kind of graph is guaranteed to have NO odd cycles? A **Bipartite** graph!
- **Hint 2:** To build a bipartite graph, split your 7 vertices into two distinct groups (a Left group and a Right group).
- **Hint 3:** Remember, edges can only go between the Left group and the Right group.
- **Hint 4:** Try splitting the 7 vertices into a group of 3 and a group of 4. How many edges can you draw between them if you connect every vertex on the Left to every vertex on the Right? Is it enough to reach 10 edges?

Further Reading & Practice

If you want to dive deeper into Olympiad and Algorithmic Graph Theory:

- **Olympiad Graph Theory** by Adam Kelly
- **Math 179: Graph Theory** by Evan Chen
- **IMO Training: Graph Theory** by Yufei Zhao
- **Graph Theory Concepts for Olympiads** (INMO Camp Notes)